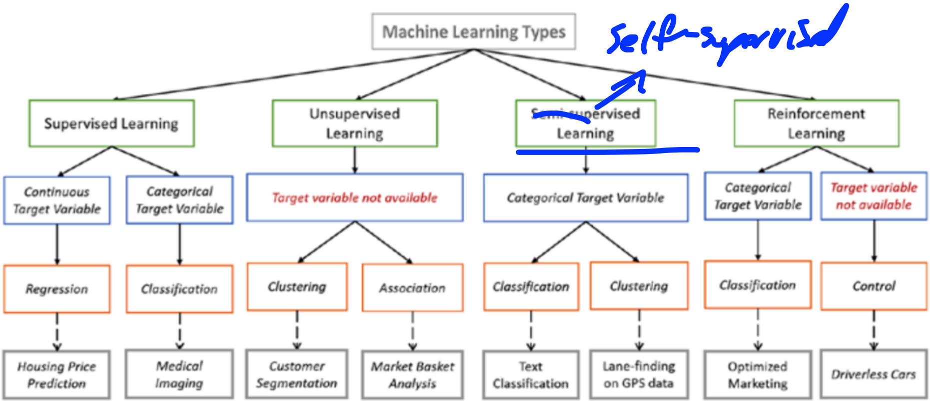
Deep Learning : Autoencoders

Mark Crowley

Outline

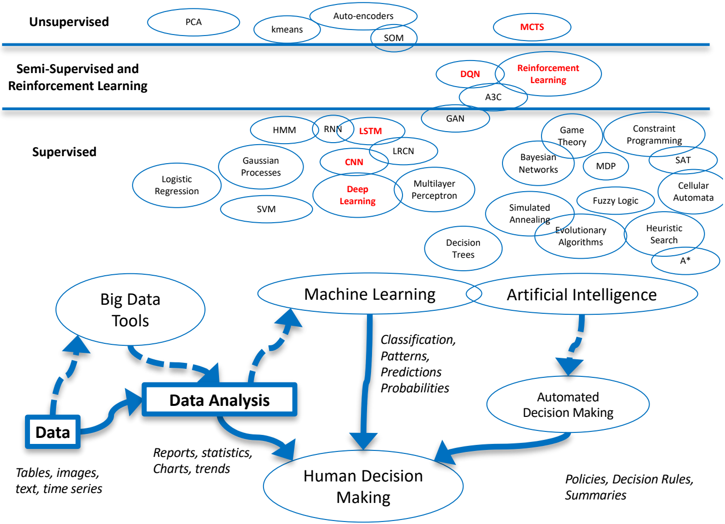
1. Super-,Self, Semi-,Un-Supervised and all that…
2. Basic Encoder-Decoders
3. Denoising Autoencoders
4. Examples and Future Directions
5. Variational Autoencoders (see online video)

Dividing Up Machine Learning



-- “Machine Learning: A Probabilistic Approach”, Kevin Murphy

Dividing Up Machine Learning



Semi-supervised or Self-supervised?

the right inductive bias for the data

In self-supervised learning, the system learns to predict part of its input from other parts of it input. In other words, a portion of the input is used as a supervisory signal to a predictor fed with the remaining portion of the input.

-- Yann LeCun

https://ai.facebook.com/blog/self-supervised-learning-the-dark-matter-of-intelligence/

Semi-supervised or Self-supervised?

"Self-supervised learning uses way more supervisory signals than supervised learning, and enormously more than reinforcement learning. That's why calling it "unsupervised" is totally misleading. That's also why more knowledge about the structure of the world can be learned through self-supervised learning than from the other two paradigms:
***the data is unlimited, and amount of feedback provided by each example is huge**."*

-- Yan LeCun

Amount of Supervisory Signal

Amount of Supervision	Method	Source of Signal	Algorithms
Very High	Self-Supervised	Input Data	Auto-Encoders, Transformers, BERT, Word2Vec
High	Supervised Learning	The Modeller/User	Logistic Regression, SVM, Random Forests, DNNs, CNNs
Moderate	Semi-Supervised	The Modeller/User	What does it mean really? Some
Low	Reinforcement Learning	The Environment the system is interacting with	
None	Unsupervised Learning?	No one?	Traditionally: PCA, Autoencoders Misleading Term?

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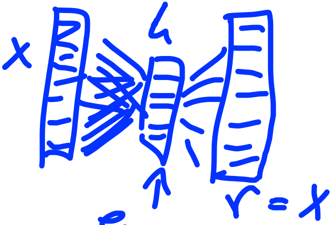
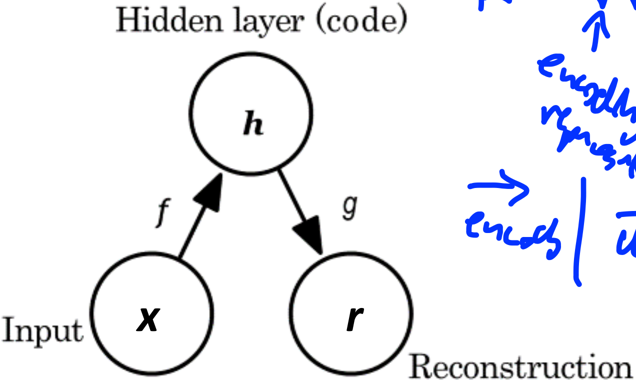
Structure of an Autoencoder

- Encoding function $h = f(x)$
- Decoding function $r = g(h)$



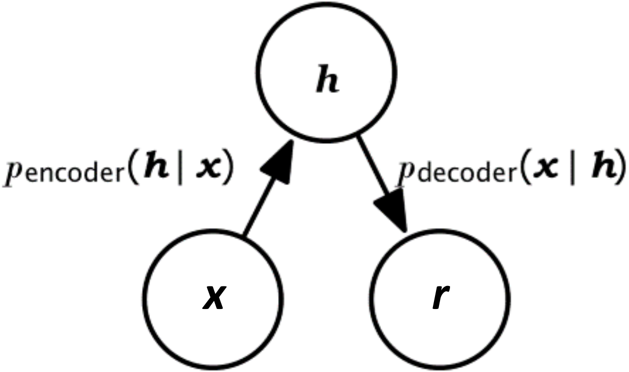
encoding representation

encoding | decoding



Stochastic Autoencoders

- Functions become probability distributions.
- Find most likely encoding *h* to explain data *x*

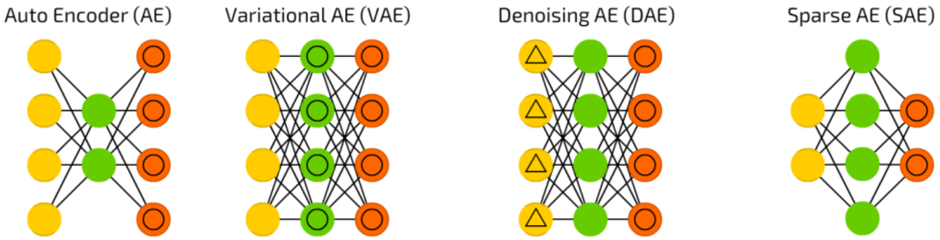


Avoiding Trivial Identity

- **Undercomplete autoencoders** $|h| < |x|$
  - *h* has lower dimension than *x*
  - *f* or *g* has low *capacity*
  - Minimize loss e.g. $L = MSE(x, g(f(x)))$
  - Simple case:
    - If decoder is linear and loss is MSE
    - Then Autoencoders learns PCA subspace! $\rightarrow |h| = |x|$
  - Nonlinear *f* and *g* can easily overfit
- **Overcomplete autoencoders** $|h| > |x|$
  - *h* has higher dimension than *x*
  - can overfit even with linear *f* and g *functions*
- **Solution:** Must be regularized

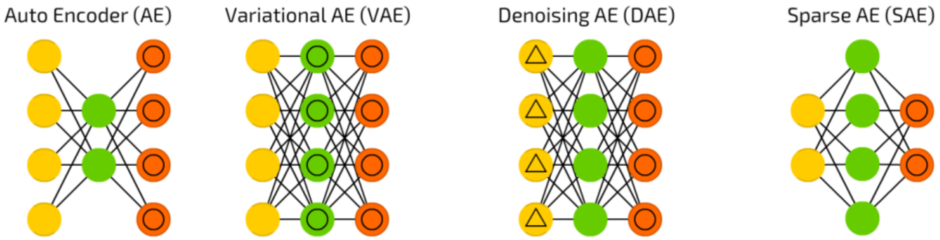
Types of Regularized Autoencoders

In addition to making the hidden layer *h* lower dimensionality to the inputs and outputs. Enforce other regularization properties:



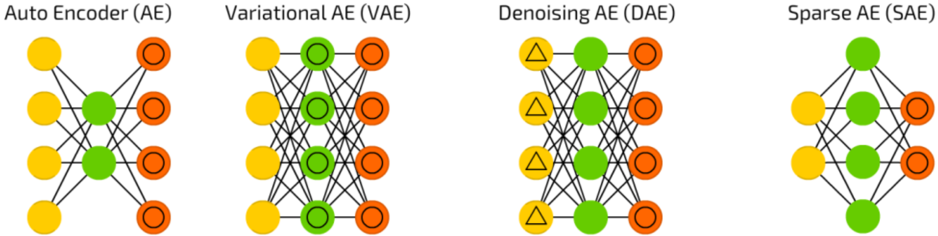
Types of Regularized Autoencoders

- **Sparse autoencoders**
  - Limit capacity of autoencoder by adding a term to the cost function penalizing the code for being larger



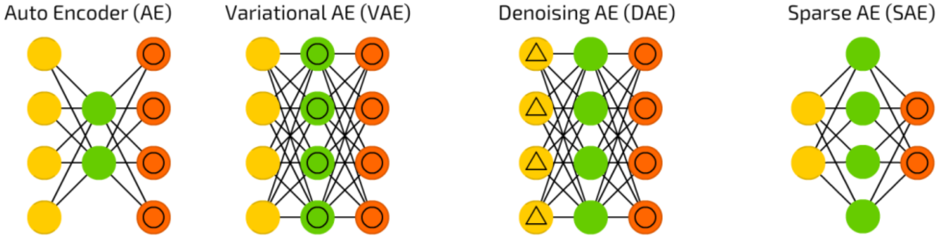
Types of Regularized Autoencoders

- **Contractive autoencoders**
  - Penalization term added to gradient – this inhibits updating the weights too quickly based on new evidence



Types of Regularized Autoencoders

- Variational Autoencoders
  - See next Lecture on Youtube
- Autoencoders with dropout on the hidden layer
- Denoising autoencoders



Difference Between Autoencoders and Sequence-to-Sequence (S2S)

enc dec

- In a **S2S** approach, the goal is to convert one sequence of inputs to a corresponding sequence of outputs.

  t-2 t-1 t t+1 t+2

- Translation from one language to another is the most often listed example.

- A simple form of this could be using the input sequence as the output sequence, also known as **Teacher Forcing**.

- This has a very similar feel to Autoencoders, but:
  - S2S is on temporal data, so LSTMs are usually used
  - The output in S2S is usually **different** from the input, or a *different version* of the input.

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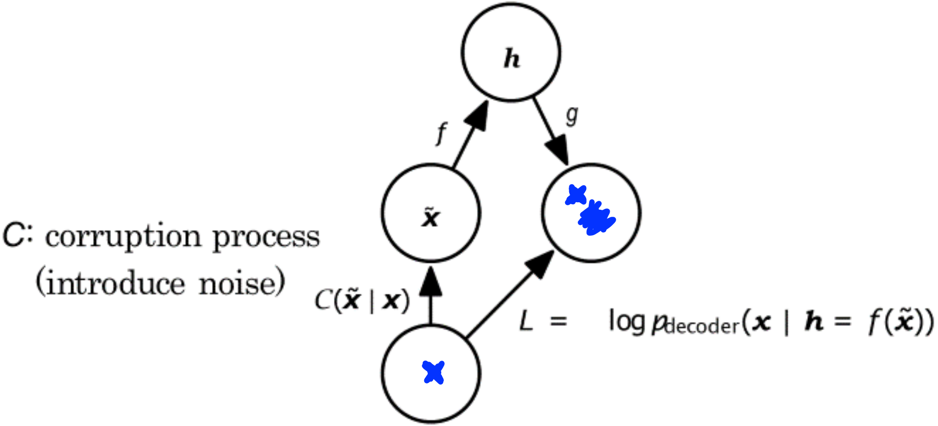
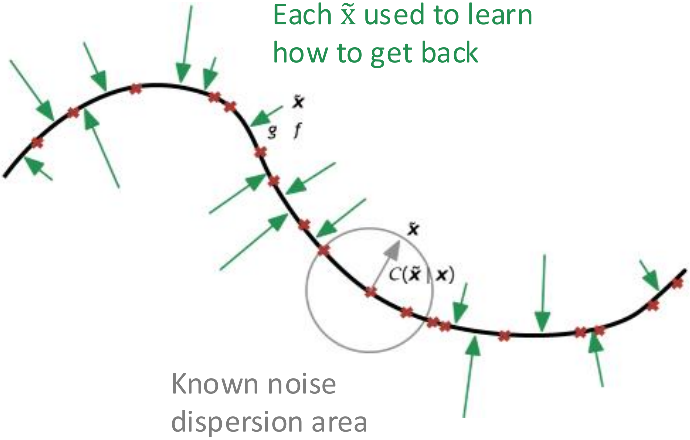
Denoising Autoencoder

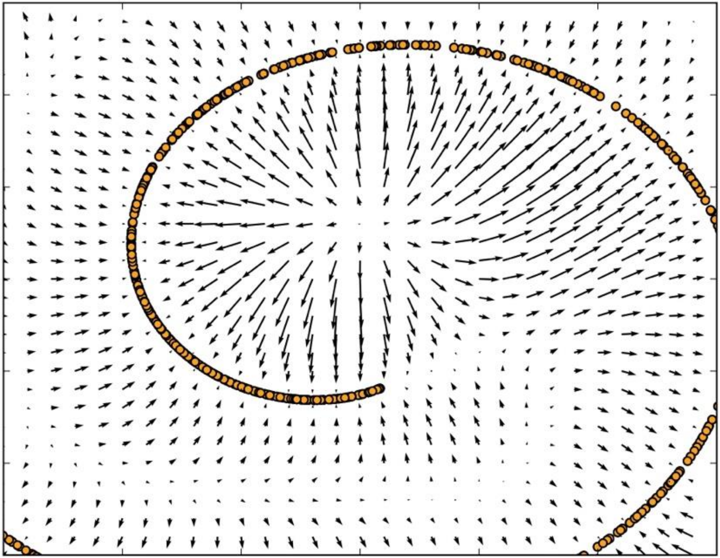
- Add noise to x to produce $\tilde{x}$. Force decoder to be compact and fix the noise.
- Can work with h being higher dimensional too



DAE Learns a Gradient Towards a Manifold



DAE Learns a Gradient Towards a Manifold



Single Hidden Layer vs. Deep Autoencoders

- Adding more hidden layers ***can***:
  - Exponentially *increases* power of autoencoder
  - Exponential *reduces* the amount of training data needed to achieve same reconstruction score
- This is not automatic, but it means deeper autoencoders often work even better:
  - Learn stacks of shallow autoencoders
  - Combine them together

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